

Supply chain management Analysis

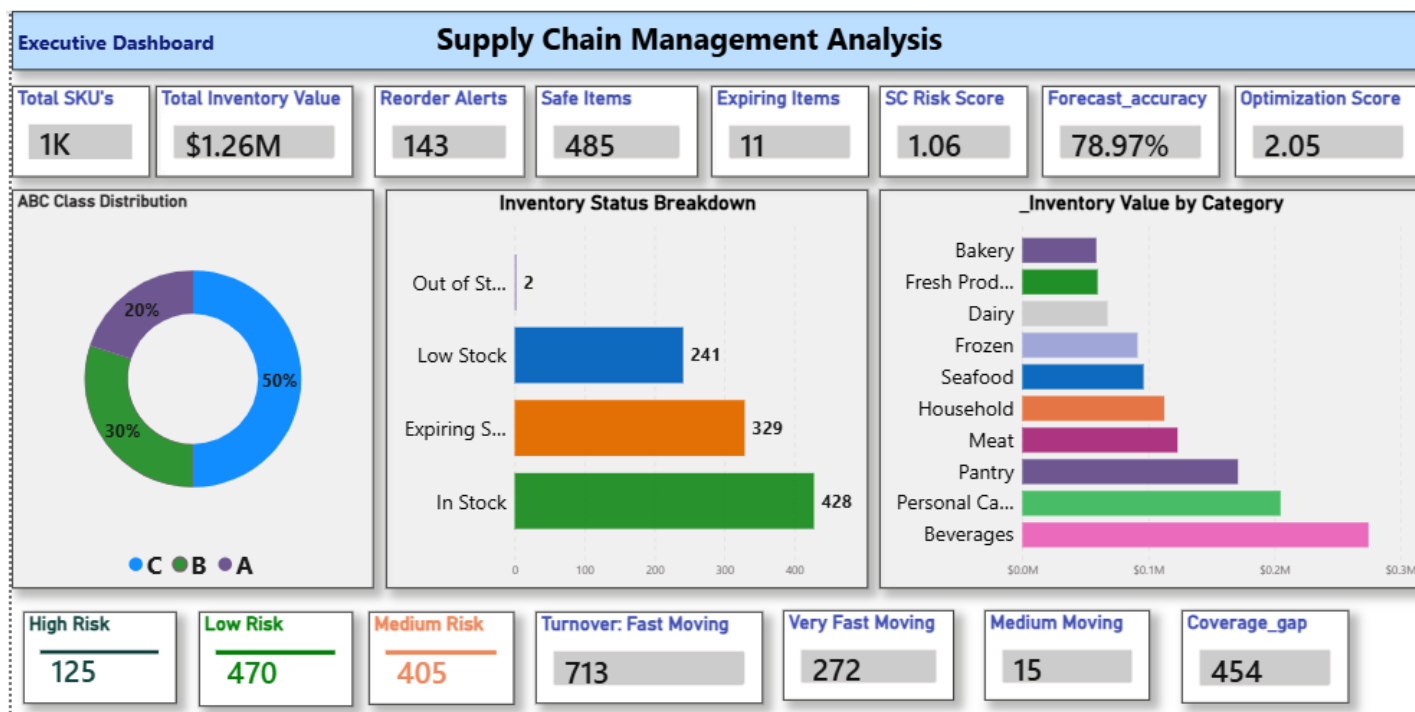
Summary:

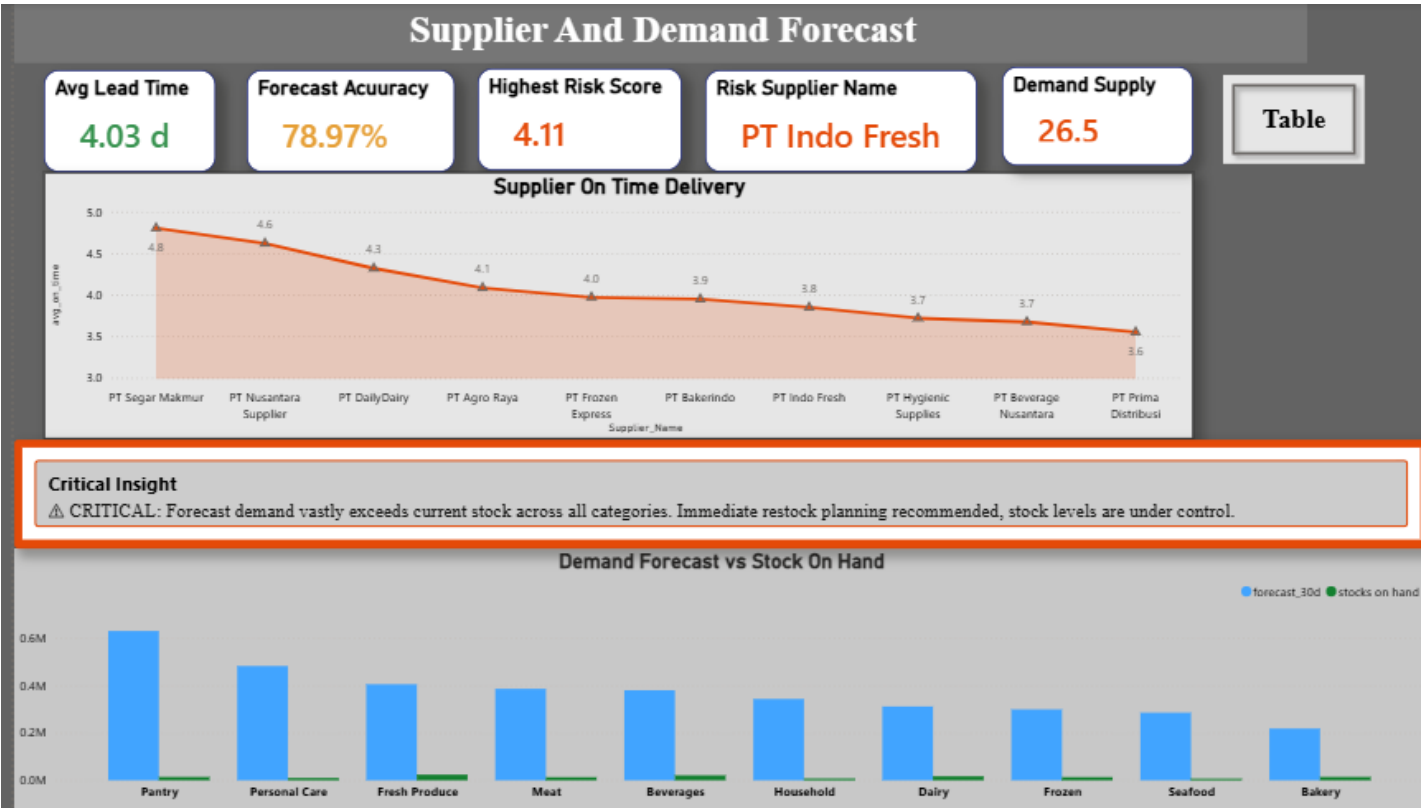
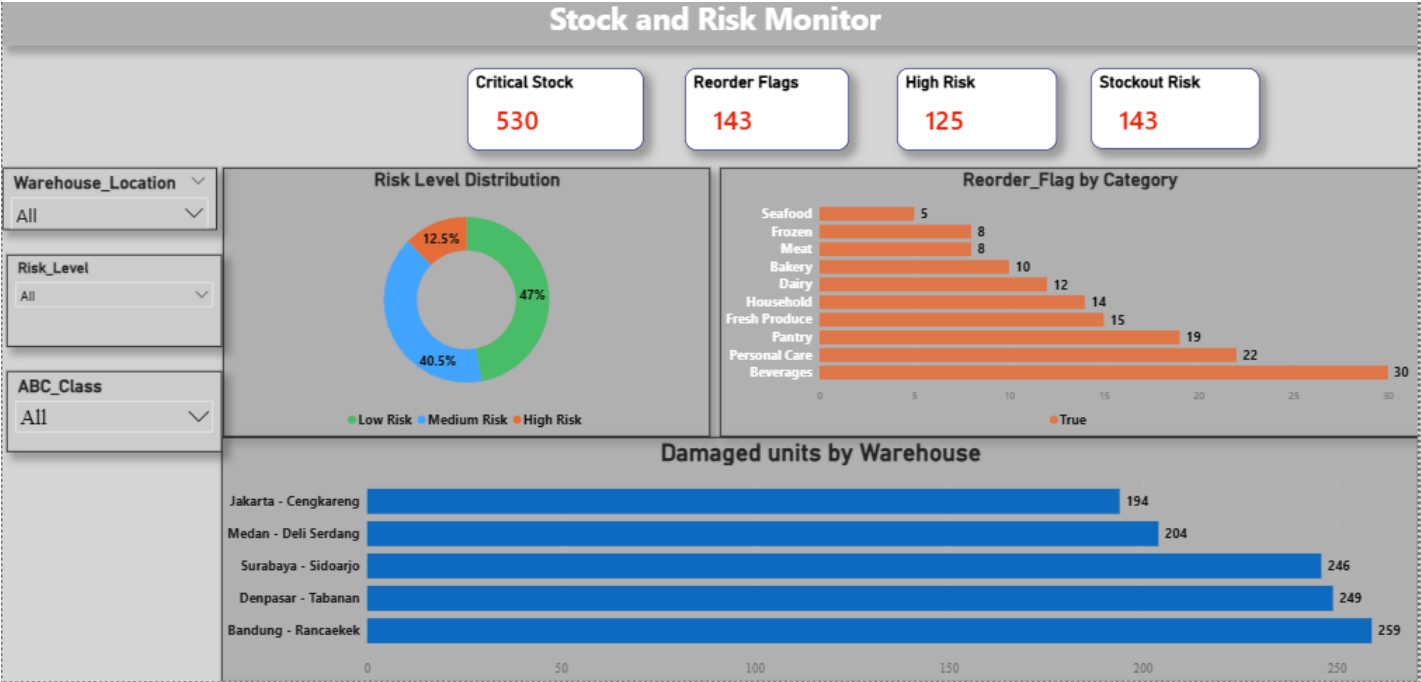
This project analyses supply chain performance across inventory, demand, suppliers, warehouses, and risk factors. The goal is to identify inefficiencies, reduce stockouts, optimize inventory levels, and improve supplier reliability.

Key outcomes:

- Identified stock imbalance issues (overstock + stockouts)
- Highlighted high-risk suppliers and delayed deliveries
- Found demand-supply mismatches across categories
- Provided data-driven reorder and optimization strategies

Dashboards:





Insights:

Inventory Insights:

- A significant portion of SKUs fall under low stock or out-of-stock categories, indicating poor replenishment planning.
- Some SKUs show excess inventory, leading to holding costs.
- Inventory Coverage Gap shows many products cannot meet demand during lead time.

Insight: Inventory is not balanced — simultaneous overstocking and understocking.

Demand Analysis:

Demand distribution is right-skewed:

- Most products have low demand
- Few products drive high demand (fast-moving SKUs)
- High-demand SKUs are more prone to stockouts

Focus should be on ABC classification:

- A → High priority
- B → Moderate
- C → Low

Insights: Implementing ABC classification helps prioritize critical products, reducing stockouts for A items while optimizing inventory costs for low-demand C items.

Supplier Performance Analysis:

- Majority of suppliers fall under Class A (Good), but a small portion shows delays.
- Some suppliers are close to risk threshold, requiring monitoring.
- Supplier On-Time % varies significantly.

Insights: Most suppliers perform reliably under Class A, but variability in on-time delivery indicates potential risks that require continuous monitoring.

Warehouse Analysis:

- Uneven stock distribution across warehouses
- Some warehouses:
 - Overstocked
 - Others facing shortages

Insights: Uneven stock distribution across warehouses leads to inefficiencies, with some locations facing overstocking costs while others risk stockouts and lost sales.

Risk Analysis:

- High stockout risk observed in fast-moving categories
- Supplier delays + low stock = critical risk zones
- Risk Score identifies vulnerable SKUs

Insights: High stockout risk is concentrated in fast-moving SKUs, especially where low inventory aligns with supplier delays, creating critical vulnerability zones.

Using Risk Score to identify these SKUs enables proactive actions like safety stock planning and supplier optimization to prevent disruptions.

Reorder Analysis:

Reorder Point defines the **exact inventory level at which a new order should be placed.**

It considers:

- Average demand
- Supplier lead time

This ensures:

- Continuous product availability
- Optimized inventory levels
- Reduced emergency ordering

Insights:

A high number of SKUs require frequent replenishment, and delays in reorder decisions increase the risk of stockouts and service disruptions.

Warehouses with higher reorder counts indicate possible under-ordering patterns or slow supplier response times.

Optimization Analysis:

Optimization Score

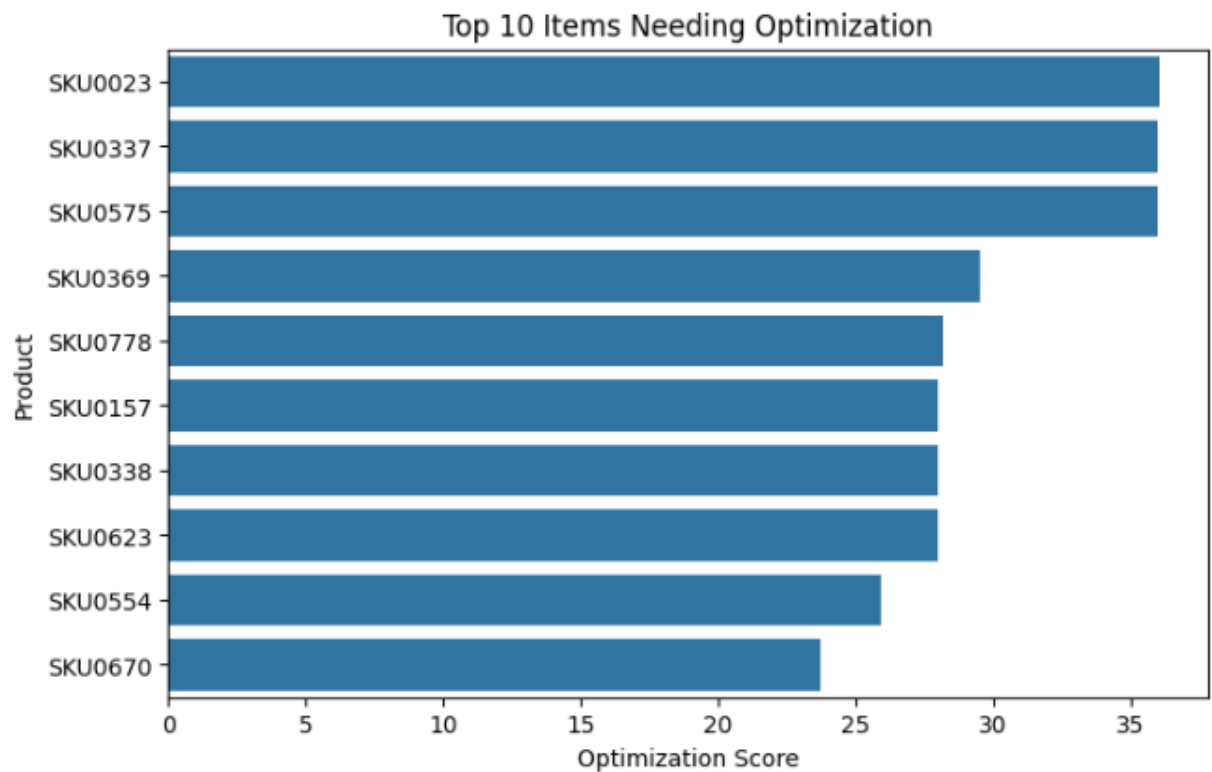
Combines:

- Risk level
- Reorder signals
- Inventory coverage

Used to identify **which SKUs need immediate action**

SKU-Level Prioritization

- The horizontal bar chart acts as a **daily action list** for inventory managers
- SKUs are ranked based on optimization score

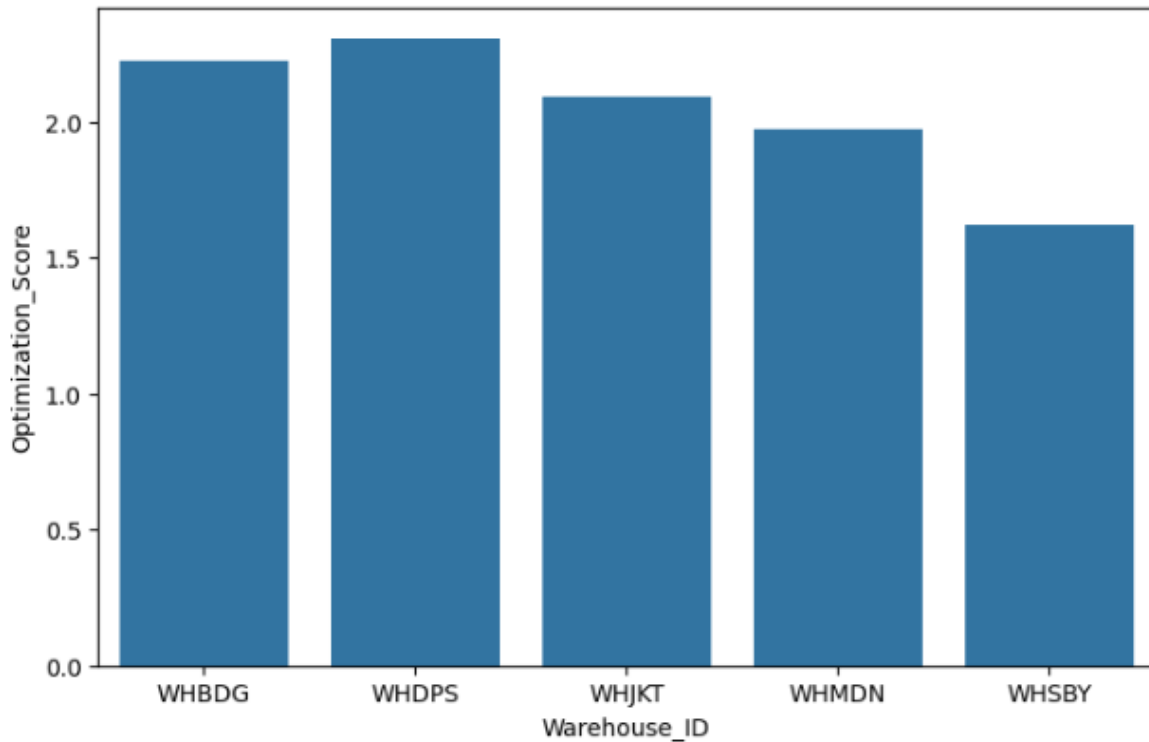


Key Rule:

- Top SKU = **highest risk item**
- Score > 15 = **requires emergency procurement**

Warehouse-Level Analysis

- Average optimization score is calculated per warehouse



Helps to identify:

This chart shows the average optimization score per warehouse, where higher scores indicate greater inefficiency and risk.

WHDPS requires the most attention, while WHSBY performs best; focusing on high-score warehouses can improve inventory efficiency and reduce risks.

Insights:

- High optimization scores indicate SKUs and warehouses with stock imbalance and higher risk.
- Top-priority SKUs need immediate action to prevent stockouts and disruptions.
- Improving high-score warehouses can enhance efficiency and reduce inventory costs.

Business Recommendations – Supply Chain Optimization

Improve Inventory Balance

- Implement **demand-driven replenishment** to avoid overstocking and stockouts
- Set **minimum and maximum stock thresholds** for each SKU
- Regularly monitor **inventory coverage gap** to ensure demand is met during lead time

Apply ABC Classification Strategy

- Focus on **A-category (high-demand) SKUs** with:
 - Higher safety stock
 - Frequent replenishment
- Reduce excess stock for **C-category items** to lower holding costs

Strengthen Supplier Performance

- Continuously track **Supplier On-Time % and risk score**
- Work closely with near-risk suppliers to improve delivery timelines
- Consider **alternate or backup suppliers** for critical SKUs

Optimize Warehouse Distribution

- Redistribute stock across warehouses to **balance inventory levels**
- Prioritize warehouses with **high stockout risk** for replenishment
- Use inter-warehouse transfers to reduce shortages

Reduce Stockout Risk

- Maintain **safety stock for fast-moving SKUs**
- Closely monitor SKUs with **high risk scores**
- Align supplier lead time with inventory planning

Enhance Reorder Strategy

- Automate **reorder point (ROP) calculations** based on demand and lead time
- Avoid delayed decisions by implementing **real-time reorder alerts**
- Review warehouses with high reorder frequency for planning gaps

Optimization Scores

- Use optimization score as a **daily action tool**
- Immediately address SKUs with **high scores (>15)**
- Focus on warehouses with higher average scores for **process improvement**

Final summary:

This supply chain analysis reveals a system struggling with **imbalance and inefficiency**, where demand is uneven, inventory is misaligned, and supplier variability adds risk. While a small set of high-demand SKUs drives business value, they are also the most vulnerable to stockouts due to delayed replenishment and supplier delays. At the same time, excess stock in low-demand items increases holding costs, and uneven warehouse distribution further amplifies inefficiencies.

By integrating demand patterns, supplier performance, warehouse distribution, and risk signals, the analysis provides a **clear, data-driven action plan**—prioritizing critical SKUs, improving reorder strategies, and optimizing inventory allocation to build a more resilient and efficient supply chain.

Conclusion

Aligning inventory with demand, strengthening supplier reliability, and acting on risk and optimization insights will transform the supply chain from reactive to proactive, ensuring better availability, lower costs, and improved operational performance.

